

Dual Representatives of Mean-Force Equilibrium as a Duality Geometry

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(Dated: February 27, 2026)

The Hamiltonian of mean force is exact but often difficult to use in practice because low-temperature and strong-coupling limits are not equally tractable in one parameter representation. We present a physical duality-geometry view in which two discrete representative maps reorganize this difficulty without changing the underlying state family. The first map is a strong-weak reflection of the crossover coordinate $y = \log \chi$, yielding $\chi \mapsto \chi^{-1}$ and, for one-channel scaling, $g^\vee = g_*^2/g$. The second map is an orientation-gauge flip of the mean-force axis that induces an affine reflection of effective-temperature representatives, $\beta'_{\text{eff}} = 2\beta - \beta_{\text{eff}}$. In the main text we emphasize physical interpretation, concrete control logic, and observable consequences; formal theorem-level statements and full proofs are collected in the Appendix. The spin-boson qubit is used as a first worked example after the general framework is established, including closure-level spectral-class thresholds and a closed entropy map. Numerical diagnostics confirm that asymptotic reconstructions are inaccurate only near crossover and rapidly become quantitatively precise deep in the target branch.

INTRODUCTION

Dualities are one of the most pervasive organizing ideas in physics: they connect weak and strong descriptions, high and low scales, and complementary parametrizations of the same underlying phenomenon. The Hamiltonian of mean force (HMF) is the exact equilibrium generator for an open quantum subsystem [1–5]. The challenge is not definition but use: once subsystem-bath coupling is finite, physically relevant limits are often controlled by different small parameters in different parts of the same phase plane. In practice this makes one branch easy and its partner hard.

In this setting, the most sought-after target is typically the ground-state sector ($\beta \rightarrow \infty$), which is often the least numerically forgiving.

This paper is built around a simple operational question: *when a target limit is hard in one representative, can we map it to an easier one while keeping the same underlying state family?* Our answer is yes, provided one works at the level of representative labels rather than parameter-level state identities.

Physical framing. Start from

$$H_{\text{tot}} = H_Q + H_X + H_{\text{int}}, \quad (1)$$

and define

$$\bar{\rho}_Q(\beta) = \text{Tr}_X e^{-\beta H_{\text{tot}}}. \quad (2)$$

The HMF is exactly the operator satisfying

$$\frac{\bar{\rho}_Q(\beta)}{Z_X(\beta)} = e^{-\beta H_{\text{MF}}(\beta)}. \quad (3)$$

The key structural move is to split this into a bare thermal factor and an influence factor. That split exposes two independent discrete freedoms:

1. a strong-weak representative reflection across crossover, and

2. an orientation-gauge choice of mean-force eigenaxis labeling.

Main message. These maps do not claim physical-state equality under inversion. They define a *duality geometry*: they exchange asymptotic work between partner representatives. In concrete terms, a low- T , strong-influence calculation can be rewritten as a weak-influence calculation at a dual representative, plus a separate orientation-gauge transformation for directed observables. This maps the numerically hard ground-state limit to a numerically trivial dual regime.

Why this viewpoint is useful. The framework turns qualitative statements like “strong coupling behaves differently” into explicit, testable map identities. It also clarifies which observables are gauge-even (radial/purity sector) and which are gauge-odd (oriented sector), which matters for fitting and collapsing data across control regimes.

Theory-first structure. The next framework sections develop representative geometry in model-independent terms. Only then do we introduce the spin-boson qubit as Example 1, where closure formulas allow direct demonstrations: closure-level spectral-class thresholds, dual-branch entropy formulas, and quantitative error diagnostics. Formal theorem-level statements and proofs are collected in the Appendix.

To avoid overclaiming, we explicitly separate closure-level representative statements from established full-model spin-boson critical phenomenology [6–9].

Relative to the closure-focused manuscript [10], the present work shifts emphasis from “deriving one exact model” to “organizing a broad class of mean-force calculations by dual representative structure” [11–13].

GENERAL DUALITY FRAMEWORK

Subsystem equilibrium with finite coupling. Consider

$$H_{\text{tot}} = H_Q + H_X + H_{\text{int}}, \quad H_{\text{int}} = g f \otimes B. \quad (4)$$

Here g is a coupling amplitude, f is a subsystem operator, and B is a bath operator. Define

$$Z_X(\beta) := \text{Tr}_X e^{-\beta H_X}, \quad Z_Q(\beta) := \text{Tr}_Q e^{-\beta H_Q}, \quad (5)$$

and the reduced equilibrium operator

$$\bar{\rho}_Q(\beta) := \text{Tr}_X e^{-\beta H_{\text{tot}}}, \quad (6)$$

with normalized reduced state

$$\rho_Q(\beta) = \frac{\bar{\rho}_Q(\beta)}{\text{Tr}_Q \bar{\rho}_Q(\beta)}. \quad (7)$$

The Hamiltonian of mean force is the exact generator satisfying

$$e^{-\beta H_{\text{MF}}(\beta)} := \frac{\bar{\rho}_Q(\beta)}{Z_X(\beta)}. \quad (8)$$

No perturbative approximation is used in this definition [1–4].

Definition 1 (Influence operator and thermal-influence partition). Define S_β by

$$e^{S_\beta} := e^{+\beta H_Q/2} \frac{\bar{\rho}_Q(\beta)}{Z_X(\beta)} e^{+\beta H_Q/2}. \quad (9)$$

Equivalently,

$$\frac{\bar{\rho}_Q(\beta)}{Z_X(\beta)} = e^{-\beta H_Q/2} e^{S_\beta} e^{-\beta H_Q/2}. \quad (10)$$

With $\rho_Q^{\text{th}} = Z_Q^{-1} e^{-\beta H_Q}$, this becomes

$$\rho_Q(\beta) = \frac{Z_Q(\beta)}{Z_{\text{MF}}(\beta)} (\rho_Q^{\text{th}}(\beta))^{1/2} e^{S_\beta} (\rho_Q^{\text{th}}(\beta))^{1/2}, \quad (11)$$

where $Z_{\text{MF}}(\beta) := \text{Tr}_Q e^{-\beta H_{\text{MF}}(\beta)}$.

Physical meaning of the partition. The bare Gibbs factor carries the reference equilibrium geometry set by H_Q , while e^{S_β} carries all coupling-induced dressing. The duality maps introduced below act on representatives of this dressed family. They do not change the underlying family itself. This thermal-plus-influence split is the operator-level equilibrium analogue of influence-functional dressing [10, 14].

Definition 2 (Representative family). Let $\mathcal{P}$ be a parameter manifold and $\mathcal{O}$ a set of observables. A representative family is a map

$$\mathcal{R} : \mathcal{P} \rightarrow \mathcal{O}, \quad p \mapsto \{O_k(p)\}_{k \in K}, \quad (12)$$

where each O_k is evaluated from one fixed analytic functional form, while p can be re-labeled by discrete representative transformations.

Crossover coordinate as influence strength. Assume there is a positive scalar χ that controls crossover in the closed solution and define

$$y := \log \chi. \quad (13)$$

The surface $\chi = 1$ (equivalently $y = 0$) is the self-dual crossover manifold separating weak ($\chi < 1$) and strong ($\chi > 1$) influence branches. For one dominant channel,

$$\chi(\beta, g, \vartheta) = g^2 \chi_0(\beta, \vartheta), \quad \chi_0 > 0, \quad (14)$$

so the moving self-dual scale is

$$g_\star(\beta, \vartheta) = \chi_0(\beta, \vartheta)^{-1/2}. \quad (15)$$

Definition 3 (Observable class used for regime transfer). An observable Q is admissible if

$$Q(\beta, g, \vartheta) = F(\beta, \vartheta; \phi_1(\chi), \dots, \phi_n(\chi), u), \quad (16)$$

where F is smooth in its arguments, each ϕ_j has asymptotic expansions for $\chi \rightarrow 0$ and $\chi \rightarrow \infty$, and u is an oriented scalar (possibly absent) that flips sign under orientation-gauge transformation.

Why this class is sufficient. If a transformation acts only on (χ, u) , then the same closed function F is reused. This is the mechanism behind regime transfer: one does not alter the analytic form, only the argument regime where it is evaluated.

Roadmap from framework to physics. The next sections introduce strong-weak representative reflection, orientation gauge, and their composition into a duality geometry. The spin-boson realization is treated only after this framework is complete.

STRONG-WEAK REPRESENTATIVE INVOLUTION

Physical idea. In many closed mean-force parameterizations, one scalar controls the crossover between two asymptotic languages: a weak-influence expansion in powers of χ , and a strong-influence expansion in powers of $1/\chi$. The key duality is therefore a reflection of the crossover coordinate,

$$y = \log \chi \mapsto -y, \quad (17)$$

which is equivalent to

$$\chi \mapsto \chi^{-1}. \quad (18)$$

Representative map at fixed β . For one-channel scaling,

$$\chi(\beta, g, \vartheta) = g^2 \chi_0(\beta, \vartheta), \quad (19)$$

define

$$g_\star(\beta, \vartheta) := \chi_0(\beta, \vartheta)^{-1/2}, \quad g^\vee := \frac{g_\star^2}{g} = \frac{1}{\chi_0 g}. \quad (20)$$

Then

$$\chi(\beta, g^\vee, \vartheta) = \chi(\beta, g, \vartheta)^{-1}, \quad (g^\vee)^\vee = g. \quad (21)$$

So each point has a partner across the self-dual contour $\chi = 1$.

Control transfer, not state equality. The dual map does *not* claim $Q(\beta, g, \vartheta) = Q(\beta, g^\vee, \vartheta)$ pointwise. Instead, it states that the hard strong-branch expansion at (β, g) can be reconstructed from the weak-branch expansion of the same functional family at $(\beta, g^\vee)$. This is a computational statement about asymptotic control.

Concrete example. For

$$\gamma(\chi) = \frac{\tanh \chi}{\chi}, \quad (22)$$

one has

$$\gamma(\chi) \sim 1 - \frac{\chi^2}{3} + O(\chi^4), \quad \chi \rightarrow 0, \quad (23)$$

$$\gamma(\chi) \sim \frac{1}{\chi}, \quad \chi \rightarrow \infty. \quad (24)$$

Under $\chi \mapsto \chi^{-1}$, the weak expansion variable at the dual representative is exactly the strong expansion variable of the original representative.

Geometric reading. In (β, g) -space, $\chi = 1$ is the self-dual manifold. The map $g \leftrightarrow g^\vee$ at fixed β is a reflection across this manifold in the coordinate $y = \log \chi$. The normal-duality construction later in the paper generalizes this to simultaneous (β, g) -motion along local normals.

Formal statement. A theorem-level formulation (including admissibility conditions and proof) is given in the Appendix.

ORIENTATION GAUGE AND AFFINE TEMPERATURE REFLECTION

Physical origin of the gauge. The mean-force generator is determined up to a discrete eigenlabel choice: which direction is called “positive” along the mean-force axis. For qubits this is a $\mathbb{Z}_2$ orientation gauge. In closed parametrizations it appears as

$$\Sigma_z \mapsto -\Sigma_z. \quad (25)$$

This flip changes oriented longitudinal information but leaves radial information unchanged.

Effective-temperature representative. Write

$$\Theta = \operatorname{arctanh}(u) - \frac{\beta \omega_q}{2}, \quad u := \gamma(\chi)U, \quad (26)$$

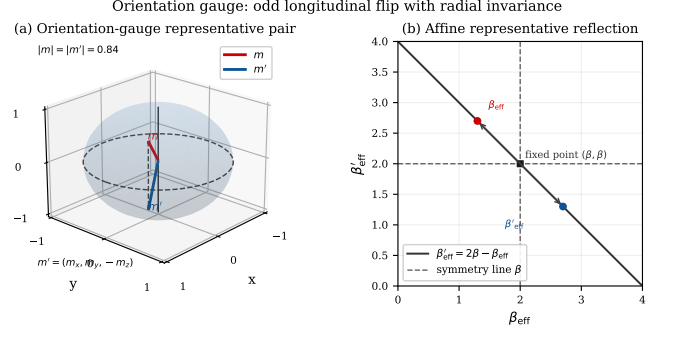


FIG. 1. Geometric action of orientation gauge. **(a)** Bloch-sphere representative pair m and $m' = (m_x, m_y, -m_z)$: radius is invariant while longitudinal orientation flips. **(b)** Effective-temperature representative reflection $\beta'_{\text{eff}} = 2\beta - \beta_{\text{eff}}$, with fixed point (β, β) .

and define

$$\beta_{\text{eff}} := -\frac{2}{\omega_q} \Theta = \beta - \delta\beta, \quad \delta\beta := \frac{2}{\omega_q} \operatorname{arctanh}(u). \quad (27)$$

Under orientation gauge, $u \mapsto -u$, hence $\delta\beta \mapsto -\delta\beta$, and therefore

$$\beta'_{\text{eff}} = 2\beta - \beta_{\text{eff}}. \quad (28)$$

So gauge partners are related by an *affine reflection* about β , not by a simple sign flip.

Even/odd observable split. Any observable depending only on χ (radius, purity, entropy from radius) is gauge-even. Observables that depend on oriented scalars are gauge-odd or affine-transformed. This split is central when comparing fitted parameters extracted from directional observables.

Practical consequence. When two data points differ only by orientation gauge, radial diagnostics should collapse directly, while oriented diagnostics should match only after the affine map $\beta_{\text{eff}} \leftrightarrow 2\beta - \beta_{\text{eff}}$ is applied.

Formal statement. Lemma-level statements and short proofs are collected in the Appendix.

COMPOSED DUALITY ACTION AND OBSERVABLE CONSTRAINTS

Two independent reflections. The strong-weak map acts on crossover strength χ , while orientation gauge acts on directed labels. Together they define a composed representative action

$$\mathcal{D}_{\text{comp}} := \mathcal{G} \circ \mathcal{D}_{\text{sw}}. \quad (29)$$

On radial observables these operations commute trivially because orientation leaves χ unchanged. On oriented observables they commute when orientation parity is preserved under strong-weak mapping.

Constraint for effective-temperature collapse. Suppose a saturated regime is parameterized as $r = r(\beta_{\text{eff}})$. Orientation pairing then requires

$$r' = r(2\beta - \beta_{\text{eff}}), \quad (30)$$

so any empirical collapse in terms of β_{eff} must satisfy the diagnostic ratio

$$R(\beta, \vartheta) := \frac{r(2\beta - \beta_{\text{eff}})}{r(\beta_{\text{eff}})}. \quad (31)$$

This provides a direct consistency check for representative fits in directional sectors.

Classical-to-quantum control exchange. Assume

$$\chi(\beta, g, \theta) = g^2 \chi_0(\beta, \theta), \quad (32)$$

and endpoint behavior

$$\chi_0(\beta, \theta) \rightarrow 0 \ (\beta \rightarrow 0), \quad \chi_0(\beta, \theta) \rightarrow \infty \ (\beta \rightarrow \infty). \quad (33)$$

At fixed g , this implies

$$\beta \rightarrow 0 : \chi \rightarrow 0, \quad g^\vee \rightarrow \infty, \quad (34)$$

$$\beta \rightarrow \infty : \chi \rightarrow \infty, \quad g^\vee \rightarrow 0. \quad (35)$$

So high-temperature weak influence and low-temperature strong influence are dual control sectors. This exchange is computationally useful because asymptotics can be evaluated on whichever branch is better conditioned.

Interpretive summary. The composed map is best viewed as a bookkeeping device that preserves physics while changing computational viewpoint:

1. strong-weak map exchanges the radial control branch,
2. orientation gauge aligns directed representatives,
3. composed action enables branch transfer with explicit parity tracking.

Formal statements. Group-structure and involution formulations are provided in the Appendix.

NORMAL DUALITY AND GEOMETRIZATION

From fixed- β reflection to geometric reflection. The map $g \mapsto g^\vee$ at fixed β already reflects the scalar coordinate $y = \log \chi$. A natural extension is to ask whether both β and g can change simultaneously so that partner points are connected by a direction normal to the crossover curve.

Crossover manifold as a geometric object. At fixed θ , define

$$y(\beta, g, \theta) = \log \chi(\beta, g, \theta), \quad \mathcal{C}_\theta := \{(\beta, g) : y = 0\}. \quad (36)$$

In one-channel scaling,

$$y = 2 \log g + \log \chi_0(\beta, \theta), \quad \nabla y = \left(\partial_\beta \log \chi_0, \frac{2}{g} \right). \quad (37)$$

So ∇y defines the local normal direction to the self-dual contour.

Normal dual representative. Given a point $p = (\beta, g)$ near $\mathcal{C}_\theta$, project it to the closest crossover point $\pi(p)$ and follow the local normal ray through $\pi(p)$. The normal-dual partner $p^\perp = (\beta^\perp, g^\perp)$ is defined as the point on the same ray satisfying

$$y(p^\perp) = -y(p). \quad (38)$$

Thus normal duality is the geometric analogue of strong-weak inversion, now in two coordinates.

What this adds physically. The fixed- β involution is recovered when the normal is nearly vertical. Away from that limit, normal duality encodes how temperature and coupling should be co-varied to keep branch distance symmetric around crossover. This is useful when experiments or numerics sweep both controls simultaneously.

Practical construction recipe. For numerical implementation:

1. compute the crossover contour $\mathcal{C}_\theta$ from $\chi = 1$,
2. estimate local normal $\hat{\nu} \propto \nabla y$,
3. for each point p , solve along the normal ray for $p^\perp$ with $y(p^\perp) = -y(p)$.

This produces a simultaneous (β, g) -dual map without introducing any additional model assumption beyond smoothness of y .

Formal statement. A local existence/involution theorem for this map is stated and proved in the Appendix.

SPIN-BOSON MODEL AS THE FIRST WORKED EXAMPLE

Role of this section. The general duality framework is model-independent. The spin-boson qubit is introduced here as the first concrete realization because it is both physically canonical and analytically closed [6, 15]. The objective is not to claim universality from one model, but to show how the representative maps become operational with explicit formulas.

From Hamiltonian to closed mean-force coordinates

We take

$$H_Q = \frac{\omega_q}{2} \sigma_z, \quad (39)$$

$$H_X = \sum_k \omega_k b_k^\dagger b_k, \quad (40)$$

$$f(\theta) = \cos \theta \sigma_z - \sin \theta \sigma_x, \quad (41)$$

$$B = \sum_k c_k (b_k + b_k^\dagger), \quad (42)$$

$$H_{\text{int}} = g f(\theta) \otimes B, \quad (43)$$

so $H_{\text{tot}} = H_Q + H_X + H_{\text{int}}$. Using Definition 1,

$$\frac{\bar{\rho}_Q(\beta)}{Z_X(\beta)} = e^{-\beta H_Q/2} e^{S_\beta} e^{-\beta H_Q/2}. \quad (44)$$

For Gaussian baths, closure compresses bath dependence into two real response scalars, $\Sigma_z(\beta, g)$ and $\Sigma_\perp(\beta, g)$ [10, 15, 16]. Define $c = \cos \theta$, $s = \sin \theta$, and

$$\chi := s \sqrt{c^2 \Sigma_\perp^2 + s^2 \Sigma_z^2}, \quad (45)$$

$$\gamma(\chi) := \frac{\tanh \chi}{\chi}, \quad (46)$$

$$\Theta := \text{arctanh}(\gamma(\chi) s^2 \Sigma_z) - \frac{\beta \omega_q}{2}, \quad (47)$$

$$m_z := \tanh \Theta, \quad r := |\tanh \chi|, \quad (48)$$

$$\tan \eta := -\cot \theta \frac{\Sigma_\perp}{\Sigma_z}. \quad (49)$$

This split has immediate physical meaning: χ controls radius/purity (how mixed the reduced state is), while (Θ, η) control orientation (where the Bloch vector points).

With one-channel scaling $K(u) = g^2 K_0(u)$,

$$\Sigma_z = g^2 \Sigma_z^{(0)}, \quad \Sigma_\perp = g^2 \Sigma_\perp^{(0)}, \quad (50)$$

so

$$\chi(\beta, g, \theta) = g^2 \chi_0(\beta, \theta), \quad (51)$$

$$g_\star(\beta, \theta) = \chi_0^{-1/2}, \quad (52)$$

$$g^\vee = \frac{g_\star^2}{g}. \quad (53)$$

Why the strong-coupling limit is analytically easy

Write the influence generator as

$$S = \Delta_0 \mathbb{I} + \Delta_z \sigma_z + \Delta_\perp (\sigma_x \cos \phi_f + \sigma_y \sin \phi_f), \quad (54)$$

and let $M = S - \Delta_0 \mathbb{I}$, so $M^2 = \chi^2 \mathbb{I}$. Then exactly

$$e^S = e^{\Delta_0} \cosh \chi (\mathbb{I} + \gamma M), \quad \gamma = \frac{\tanh \chi}{\chi}. \quad (55)$$

This closed form is the main practical simplification: in the strong sector $\chi \gg 1$, one reads off asymptotics directly from elementary functions without extra resummation. The radial saturation $r \rightarrow 1$ follows immediately from $\tanh \chi \rightarrow 1$.

Dual transfer to low-temperature control

At fixed g , low-temperature analysis can be difficult because $\chi_0(\beta, \theta)$ develops strong structure as $\beta \rightarrow \infty$. If $\chi_0 \rightarrow \infty$, then

$$g^\vee = \frac{1}{\chi_0 g} \rightarrow 0, \quad \chi(\beta, g^\vee, \theta) = \chi(\beta, g, \theta)^{-1} \rightarrow 0. \quad (56)$$

So the hard strong- χ sector at (β, g) is evaluated through the weak- χ expansion of the dual representative $(\beta, g^\vee)$. This is the central control payoff of the dual map.

Canonical threshold and angle-resolved phase structure

To obtain closure-exact critical couplings, first specialize to the canonical channel $\theta = \pi/2$ (so $c = 0$, $s = 1$). Then

$$\chi(\beta, g) = g^2 \Sigma_z^{(0)}(\beta; s), \quad \Theta(\beta, g) = \chi(\beta, g) - \frac{\beta \omega_q}{2}, \quad (57)$$

which gives

$$m_z(\beta, g; s) = \tanh \left[g^2 \Sigma_z^{(0)}(\beta; s) - \frac{\beta \omega_q}{2} \right]. \quad (58)$$

Define

$$\lambda_s := \lim_{\beta \rightarrow \infty} \frac{\Sigma_z^{(0)}(\beta; s)}{\beta}. \quad (59)$$

When $\lambda_s \in (0, \infty)$, the $T = 0$ order parameter is

$$m_0(g; s) = \text{sgn} \left(g^2 \lambda_s - \frac{\omega_q}{2} \right), \quad (60)$$

with exact critical coupling

$$g_c^2(s) = \frac{\omega_q}{2\lambda_s}. \quad (61)$$

For

$$J_s(\omega) = 2g^2 \omega_c^{1-s} \omega^s e^{-\omega/\omega_c}, \quad s > 0, \quad (62)$$

one gets

$$\lambda_s = 2\omega_c \Gamma(s), \quad g_c^2(s) = \frac{\omega_q}{4\omega_c \Gamma(s)}. \quad (63)$$

This spectral dependence is exact for the canonical $\theta = \pi/2$ slice of the closure model.

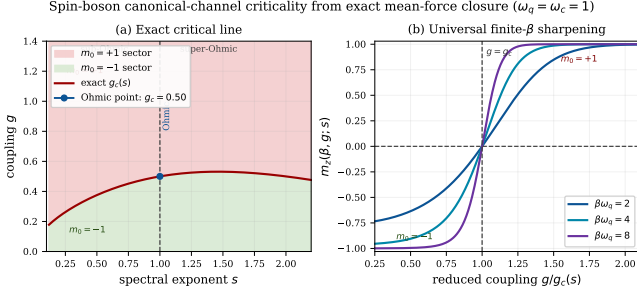


FIG. 2. Spin-boson closure criticality on the canonical channel. **(a)** Exact spectral-class critical line $g_c(s) = \sqrt{\omega_q/(4\omega_c\Gamma(s))}$. **(b)** Universal finite- β sharpening in reduced coupling $x = g/g_c(s)$, with transition pinned at $x = 1$.

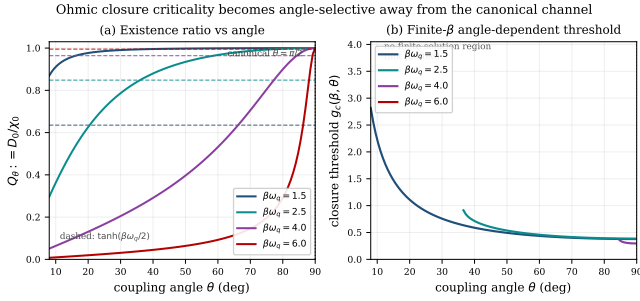


FIG. 3. Ohmic ($s = 1$) angle dependence of the closure threshold. **(a)** Existence ratio $Q_\theta(\beta)$ compared with $\tanh(\beta\omega_q/2)$ (dashed). **(b)** Finite- β $g_c(\beta, \theta)$ from Eq. (67); points with no finite solution are omitted. The threshold corridor narrows toward $\theta = \pi/2$ as temperature is lowered.

Angle dependence from the same closed equations. For generic θ , define

$$D_0(\beta, \theta) := s^2 \Sigma_z^{(0)}(\beta, \theta), \quad X_0(\beta, \theta) := \chi_0(\beta, \theta), \quad (64)$$

and

$$Q_\theta(\beta) := \frac{D_0(\beta, \theta)}{X_0(\beta, \theta)} \in (0, 1]. \quad (65)$$

The sign-change condition $m_z = 0$ ($\Theta = 0$) becomes

$$Q_\theta(\beta) \tanh(g_c^2 X_0) = \tanh\left(\frac{\beta\omega_q}{2}\right), \quad (66)$$

so

$$g_c^2(\beta, \theta) = \frac{1}{X_0} \operatorname{arctanh}\left[\frac{\tanh(\beta\omega_q/2)}{Q_\theta(\beta)}\right], \quad (67)$$

only if $\tanh(\beta\omega_q/2) < Q_\theta(\beta)$; otherwise there is no finite g_c . As $\beta \rightarrow \infty$, a finite threshold requires $Q_\theta \rightarrow 1$, i.e. the transverse sector must vanish. In practice, the finite-threshold region contracts toward the canonical axis.

Interpretive note. The phase line $g_c(s)$ is a zero-temperature closure boundary. It is distinct from the

finite-temperature crossover line $g_*(\beta, \theta)$ from $\chi = 1$. The former separates asymptotic sectors of the canonical channel; the latter controls branch exchange within finite- β calculations.

Scope relative to full spin-boson criticality. These g_c expressions are closure-level representative thresholds. They are not a replacement for full spin-boson universality results. For the unbiased model, the established phenomenology includes Ohmic KT criticality and no localization transition in the super-Ohmic case [6, 7]. For sub-Ohmic dissipation, nonperturbative NRG/QMC and analytic studies report a genuine transition with s -dependent critical behavior [8, 9, 17–19]. The present angle-resolved map is therefore a duality-geometry statement inside the mean-force closure, to be benchmarked against those full-model results rather than identified with them.

Entropy geometry and dual representative map

For any qubit state of Bloch radius r , $\lambda_\pm = (1 \pm r)/2$, so

$$S_Q = - \sum_{\pm} \lambda_{\pm} \log \lambda_{\pm}. \quad (68)$$

Using $r = |\tanh \chi|$,

$$S_Q(\beta, g, \theta) = h_2\left(\frac{1 + |\tanh \chi(\beta, g, \theta)|}{2}\right), \quad (69)$$

where h_2 is the binary entropy. Under $g \mapsto g^\vee$, $\chi \mapsto \chi^{-1}$, hence

$$S_Q(\beta, g^\vee, \theta) = h_2\left(\frac{1 + |\tanh(\chi^{-1})|}{2}\right). \quad (70)$$

This is an exact analytic map between partner representatives.

Asymptotically,

$$S_Q(\chi) = \log 2 - \frac{\chi^2}{2} + O(\chi^4), \quad \chi \rightarrow 0, \quad (71)$$

$$S_Q(\chi) = (1 + 2\chi)e^{-2\chi} + O(e^{-4\chi}), \quad \chi \rightarrow \infty. \quad (72)$$

So strong-branch entropy can be evaluated as a weak-branch expansion in $\chi^\vee = \chi^{-1}$.

Connection to earlier spin-boson literature. The trends above are consistent with analytical and numerical studies of spin-boson entanglement and localization diagnostics [20–24]. The distinct contribution here is the closed representative map that links the two branches by $\chi^\vee = \chi^{-1}$ in one analytic framework.

Formal statements and proofs. The theorem-level forms of the strong-weak and orientation results used in this section are stated in the Appendix.

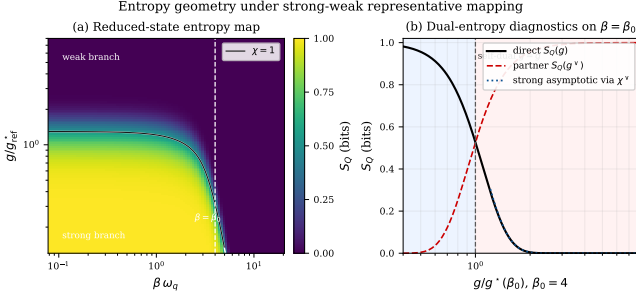


FIG. 4. Entropy under strong-weak representative mapping. (a) Reduced-state entropy over (β, g) , with crossover contour $\chi = 1$. (b) At fixed β_0 , direct entropy $S_Q(g)$, dual-partner entropy $S_Q(g^*)$, and strong-branch asymptotic reconstruction in the dual coordinate χ^* .

TABLE I. Agreement windows for $x \geq x_{\min}$. $\delta_\gamma := |\gamma_{\text{asym}} - \gamma_{\text{exact}}|/\gamma_{\text{exact}}$, $\Delta S_Q := S_Q^{\text{asym}} - S_Q^{\text{exact}}$. Entries for δ_γ are percentages; entries for $|\Delta S_Q|$ are bits.

$x_{\min}$	$\max \delta_\gamma$	$\text{RMS } \delta_\gamma$	$\max \Delta S_Q $	$\text{RMS } \Delta S_Q $
1.0	31.30%	6.68%	5.87×10^{-2}	1.08×10^{-2}
1.2	11.87%	2.33%	1.45×10^{-2}	2.30×10^{-3}
1.5	2.24%	0.389%	8.79×10^{-4}	1.16×10^{-4}
2.0	0.0668%	0.00978%	1.37×10^{-6}	1.48×10^{-7}

VALIDATION STRATEGY AND QUANTITATIVE ACCURACY

Validation philosophy. The goal is targeted falsification of the representative maps, not broad numerical benchmarking. Each diagnostic is tied directly to one physical claim: (i) involutive branch exchange, (ii) orientation-gauge affine reflection, and (iii) quantitative usefulness of asymptotic reconstruction away from crossover.

Core branch-exchange visualization

How accurate is the analytic reconstruction?

Interpretation. The asymptotic reconstruction is intentionally not expected to be accurate exactly at self-duality $x \approx 1$. What matters for control transfer is performance in the target branch. By $x \geq 1.5$, γ errors are already at the few-percent level and entropy errors are sub-millibit. By $x \geq 2$, the reconstruction is effectively exact at manuscript resolution.

Minimal acceptance checks

1. Involution: $(g^\vee)^\vee = g$ and $\chi(\beta, g^\vee, \theta) = 1/\chi(\beta, g, \theta)$.
2. Orientation: $\beta'_{\text{eff}} + \beta_{\text{eff}} = 2\beta$, with radial invariants unchanged.
3. Asymptotic transfer: direct strong-branch evaluation agrees with dual weak-branch reconstruction in the intended window.
4. Manuscript consistency: figure scripts, table values, and text-level claims are synchronized.

Reproducibility note. Core and comparison assets are produced by the project validation scripts. The quantitative table values are taken from the generated metrics CSV in the same validation pipeline.

DISCUSSION AND OUTLOOK

What this work contributes. The central contribution is a physically interpretable duality geometry for mean-force equilibrium: one reflection exchanges weak and strong influence representations, and a second reflection handles orientation labeling. Together they convert difficult asymptotic sectors into tractable partner calculations while keeping the same analytic family.

Why this is useful beyond one model. The framework only needs two structural ingredients:

1. a positive crossover coordinate with distinct weak/strong asymptotics, and
2. a closed representative map on control labels.

Whenever these exist, one can separate physical content (state family) from computational strategy (representative choice). This is a practical advantage for low-temperature and strong-coupling studies where direct expansions are fragile.

Scope and limits. The dual maps are statements about representative control, not about parameter-level equality of physical states. The spin-boson analysis provides an explicit existence example with closure-level exact formulas, including spectral-class critical couplings and entropy branch transfer. These results should be interpreted at the level of the closed mean-force representation used here; broader universality claims require model-by-model analysis.

Relation to established literature. For spin-boson criticality, the benchmark remains nonperturbative universality analyses: Ohmic KT behavior and super-Ohmic delocalization from the classic dissipative two-state framework, and sub-Ohmic criticality from NRG/QMC and related approaches [6–9, 17, 19]. For mean-force

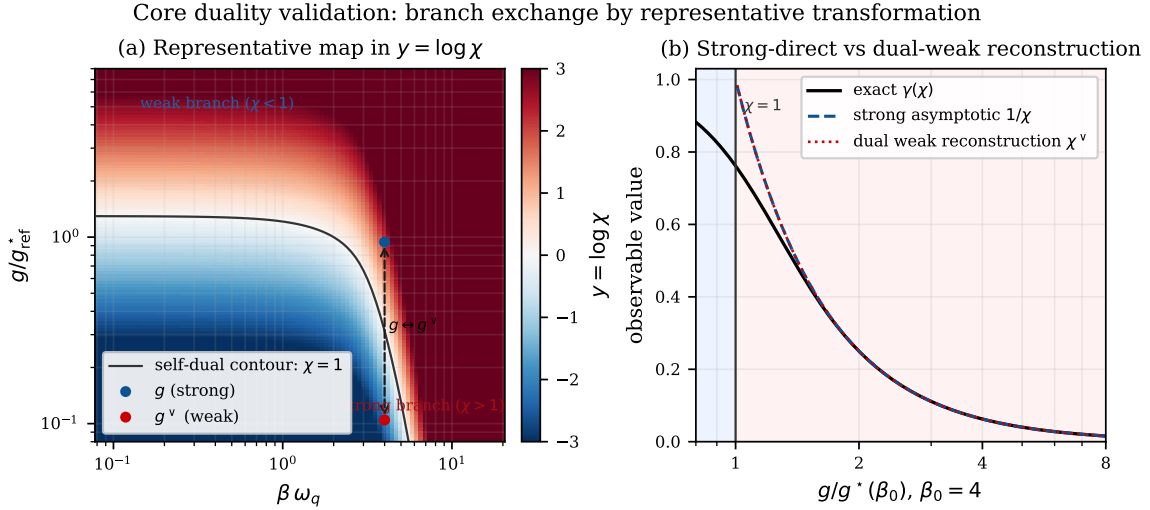


FIG. 5. Core dual-representative validation. **(a)** Map of $y = \log \chi$ in (β, g) , with self-dual contour $\chi = 1$, branch labels, and a fixed- β pair $g \leftrightarrow g^\vee$. **(b)** At fixed β_0 , direct $\gamma(\chi)$, strong asymptotic $1/\chi$, and dual weak variable $\chi^\vee$. Their convergence on $g/g^* > 1$ visualizes branch-transfer control.

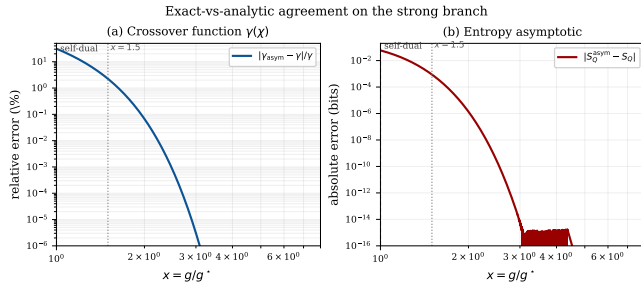


FIG. 6. Exact-vs-analytic error diagnostics on the strong branch $x := g/g^* \geq 1$. **(a)** Relative error of $\gamma_{\text{asym}} = 1/\chi$ versus $\gamma_{\text{exact}} = \tanh \chi / \chi$ with $\chi = x^2$. **(b)** Absolute entropy error in bits for $S_Q^{\text{asym}} = ((1 + 2\chi)e^{-2\chi}) / \ln 2$ versus $S_Q^{\text{exact}} = h_2((1 + |\tanh \chi|)/2)$. The dominant mismatch is confined to the immediate crossover neighborhood.

thermodynamics, the present duality-geometry perspective complements rather than replaces existing structural and operational formulations [1–5].

Experimental and numerical relevance. The geometric framing suggests straightforward diagnostics: map data into $y = \log \chi$, identify self-dual contour $\chi = 1$, and test partner reconstructions across branches. The orientation-gauge affine reflection gives an additional filter for distinguishing radial invariants from direction-sensitive observables in fitted effective descriptions.

Natural extensions. Three directions are immediate:

1. multi-channel couplings with several χ_a , where $(\mathbb{Z}_2)^k$ branch reflections may emerge,
2. finite- N systems with richer eigenlabel symmetries than qubit $\mathbb{Z}_2$, and

3. thermodynamic-cycle settings where representative transfer can reduce control-cost modeling in strong-coupling operation windows.

Bottom line. Mean-force duality is most useful when treated as a control language: not a hidden symmetry of parameters, but a map between computationally complementary views of the same dressed equilibrium state.

Formal Statements and Proofs

This appendix collects theorem-level formulations used in the main text. The body of the manuscript keeps the same results in interpretation-first form.

Strong-weak involution and asymptotic transfer

Theorem 1 (Strong-weak representative involution). *Assume one-channel scaling*

$$\chi(\beta, g, \vartheta) = g^2 \chi_0(\beta, \vartheta), \quad \chi_0(\beta, \vartheta) > 0. \quad (73)$$

Define

$$g_*(\beta, \vartheta) := \chi_0(\beta, \vartheta)^{-1/2}, \quad g^\vee := \frac{g_*^2}{g} = \frac{1}{\chi_0 g}. \quad (74)$$

Then

$$\chi(\beta, g^\vee, \vartheta) = \chi(\beta, g, \vartheta)^{-1}, \quad (75)$$

$$(g^\vee)^\vee = g, \quad (76)$$

$$y(\beta, g^\vee, \vartheta) = -y(\beta, g, \vartheta), \quad y := \log \chi. \quad (77)$$

Proof. Direct substitution gives

$$\chi(\beta, g^\vee, \vartheta) = \chi_0 \left(\frac{1}{\chi_0 g} \right)^2 = \frac{1}{\chi_0 g^2} = \chi(\beta, g, \vartheta)^{-1}. \quad (78)$$

Applying the map again,

$$(g^\vee)^\vee = \frac{1}{\chi_0 g^\vee} = \frac{1}{\chi_0} (\chi_0 g) = g. \quad (79)$$

Taking logarithms yields $y \mapsto -y$. $\square$

Proposition 1 (Asymptotic exchange for admissible observables). *Let Q satisfy Definition 3, and let each building block ϕ_j admit weak and strong asymptotic expansions at $\chi \rightarrow 0$ and $\chi \rightarrow \infty$. Then the strong- χ asymptotic expansion of Q at (β, g, ϑ) is obtained by evaluating the weak- χ expansion at $(\beta, g^\vee, \vartheta)$, and conversely.*

Proof. By Theorem 1, $\chi(\beta, g^\vee, \vartheta) = \chi(\beta, g, \vartheta)^{-1}$. Hence a weak expansion $\phi_j(\chi) \sim \sum_n a_{j,n} \chi^n$ at $g^\vee$ becomes $\sum_n a_{j,n} \chi^{-n}$ at g , i.e. a strong expansion. The reverse direction is identical. $\square$

Orientation gauge and affine reflection

Lemma 1 (Affine effective-temperature reflection). *Let*

$$\beta_{\text{eff}} = \beta - \delta\beta, \quad \delta\beta := \frac{2}{\omega_q} \operatorname{arctanh}(u), \quad (80)$$

with $|u| < 1$. Under orientation gauge $u \mapsto -u$,

$$\beta'_{\text{eff}} = 2\beta - \beta_{\text{eff}}. \quad (81)$$

Proof. Because $\operatorname{arctanh}$ is odd,

$$\delta\beta' = \frac{2}{\omega_q} \operatorname{arctanh}(-u) = -\delta\beta. \quad (82)$$

Therefore

$$\beta'_{\text{eff}} = \beta - \delta\beta' = \beta + \delta\beta = 2\beta - \beta_{\text{eff}}. \quad (83)$$

$\square$

Lemma 2 (Even/odd observable split). *If $O_{\text{rad}} = F(\chi)$, then orientation gauge leaves it invariant. If $O_{\text{odd}} = H(\chi, u)$ and $H(\chi, -u) = -H(\chi, u)$, then $O_{\text{odd}} \mapsto -O_{\text{odd}}$.*

Proof. Orientation gauge acts as $(\chi, u) \mapsto (\chi, -u)$. Substituting into F and H gives the parity statements. $\square$

Composed action structure

Proposition 2 (Composed map and commuting radial action). *Let $\mathcal{D}_{\text{sw}}$ denote strong-weak reflection and $\mathcal{G}$ orientation gauge. Both are involutions. On radial sectors,*

$$(\mathcal{G} \circ \mathcal{D}_{\text{sw}})\chi = (\mathcal{D}_{\text{sw}} \circ \mathcal{G})\chi = \chi^{-1}. \quad (84)$$

On oriented sectors they commute whenever strong-weak mapping preserves orientation parity.

Proof. Involution of $\mathcal{D}_{\text{sw}}$ follows from Theorem 1, and involution of $\mathcal{G}$ is immediate from sign flip. Orientation acts trivially on χ , so radial commutation follows. The parity-preserving condition is sufficient for commutation on oriented labels. $\square$

Normal-duality existence and involution

Definition 4 (Normal-dual map). Fix θ , define $y(\beta, g, \theta) = \log \chi(\beta, g, \theta)$, and $\mathcal{C}_\theta := \{(\beta, g) : y = 0\}$. Assume $y \in C^2$, $\nabla y \neq 0$ on $\mathcal{C}_\theta$, and a tubular neighborhood where each point p has unique projection $\pi(p) \in \mathcal{C}_\theta$. Writing

$$p = \pi(p) + \lambda \hat{\nu}(\pi(p)), \quad (85)$$

with $\hat{\nu}$ unit normal, define $p^\perp$ on the same normal ray by

$$y(p^\perp) = -y(p). \quad (86)$$

Theorem 2 (Local existence and involution of normal duality). *Under Definition 4, there exists $\delta > 0$ such that for $|\lambda| < \delta$:*

1. $p^\perp$ exists and is unique,
2. the map is involutive, $(p^\perp)^\perp = p$,
3. the segment connecting p and $p^\perp$ is orthogonal to $\mathcal{C}_\theta$ at $\pi(p)$.

Proof. For fixed $c \in \mathcal{C}_\theta$, define $\Phi_c(\lambda) := y(c + \lambda \hat{\nu}(c))$. Since $\Phi_c(0) = 0$ and $\Phi'_c(0) = \nabla y(c) \cdot \hat{\nu}(c) = \|\nabla y(c)\| > 0$, Φ_c is strictly monotone on a sufficiently small interval. Hence $\Phi_c(\lambda^\perp) = -\Phi_c(\lambda)$ has a unique solution. Applying the construction twice gives $\Phi_c(\lambda^{\perp\perp}) = \Phi_c(\lambda)$, so $\lambda^{\perp\perp} = \lambda$ by injectivity. Orthogonality is immediate because both points lie on one normal ray through c . $\square$

Equation bookkeeping index

Maintain one-to-one mapping with `paper_dual_hmf/notes/equation_registry.md`.

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